

DCT-Based Spread Spectrum Audio Watermarking with Chirp-Synchronized Extraction Over Acoustic Air-Gap Channels

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Abstract—Digital audio watermarking is a critical technique for copyright protection and content authentication. This paper presents a DCT-based spread spectrum audio watermarking system evaluated for robustness against acoustic air-gap transmission channels. The system employs pseudo-random noise (PN) sequence-based spread spectrum modulation in the Discrete Cosine Transform domain, combined with a linear chirp preamble for temporal synchronization. A 32-bit watermark payload is protected using repetition coding ($R = 7$), yielding 224 coded bits embedded across 4096-sample frames in DCT coefficients spanning bins 100–500 (538–2691 Hz). Evaluation comprised a digital baseline test confirming perfect extraction (BER = 0.0000, SNR = 23.09 dB), and acoustic air-gap experiments at 1, 3, and 5 feet using consumer hardware (laptop microphone, phone speaker). Acoustic experiments yielded mean BER values of 0.4000 ± 0.0809 at 1 ft, 0.4000 ± 0.1296 at 3 ft, and 0.4375 ± 0.0699 at 5 ft, with channel SNR averaging -2.06 to -2.55 dB. Chirp synchronization achieved peak-to-sidelobe ratios of 8.4–29.4 dB, confirming reliable alignment. The paper documents methodology evolution from QIM to spread spectrum embedding, identifying consumer hardware limitations as the primary barrier to acoustic recovery. These findings establish a baseline for future improvements using professional equipment and advanced error correction.

Index Terms—Audio watermarking, DCT, spread spectrum, acoustic air-gap, chirp synchronization, BER, copyright protection.

I. INTRODUCTION

A. Background and Motivation

The rapid proliferation of digital multimedia content has created unprecedented challenges for intellectual property protection. Digital audio is easily copied and distributed without authorization, causing significant economic losses [1]. Traditional encryption protects content during transmission but offers no protection once decrypted. Digital watermarking addresses this by embedding imperceptible identifying information directly into audio signals, enabling persistent ownership verification [2].

Audio watermarking must satisfy three competing requirements: imperceptibility, robustness, and capacity [3]. Among robustness challenges, acoustic transmission—where watermarked audio is played through a speaker and re-recorded by a microphone—represents one of the most demanding scenarios, known as the acoustic air-gap channel [4].

The acoustic air-gap scenario is particularly relevant in broadcast monitoring, where content identification systems

must detect watermarks from audio captured by microphones in public spaces, retail environments, or through secondary recordings. The ability to survive this transmission path would enable applications in copyright enforcement, audience measurement, and content tracking [5].

B. The Acoustic Air-Gap Challenge

When watermarked audio passes through a speaker-air-microphone path, it undergoes multiple forms of degradation: (i) digital-to-analog conversion artifacts, (ii) speaker frequency response distortion and nonlinear harmonic generation, (iii) acoustic propagation losses including distance-dependent attenuation and multipath reflections, (iv) ambient noise contamination from the recording environment, (v) microphone frequency response limitations and sensitivity constraints, and (vi) analog-to-digital conversion processing including automatic gain control (AGC) and noise cancellation algorithms [5].

Each of these degradation sources independently modifies the spectral characteristics of the audio signal. Their combined effect creates a transmission channel that is fundamentally different from digital processing attacks (compression, filtering, resampling) that traditional watermarking systems are designed to resist. The acoustic channel introduces both linear distortions (frequency response, propagation loss) and nonlinear distortions (speaker/microphone saturation, AGC dynamics), making it exceptionally challenging for watermark survival [6].

Furthermore, the temporal relationship between the original watermarked signal and the recorded signal is unknown—the recording may begin at an arbitrary time before or after playback starts. This synchronization problem must be solved before any watermark extraction can be attempted, adding another layer of complexity to the acoustic watermarking challenge [25].

C. Research Objective

This research investigates the design and empirical evaluation of a DCT-based spread spectrum audio watermarking system with chirp-synchronized extraction over acoustic air-gap channels. The primary objectives are:

- 1) To design a complete watermarking pipeline capable of embedding and extracting watermarks in the digital domain with zero errors.

- 2) To develop a reliable synchronization mechanism for acoustic recordings using chirp preamble correlation.
- 3) To empirically evaluate watermark survivability through consumer-grade hardware at varying distances (1 ft, 3 ft, 5 ft).
- 4) To document the iterative methodology evolution and identify the primary factors limiting acoustic watermark recovery.

D. Contributions

The key contributions of this paper are:

- 1) A complete DCT-based spread spectrum watermarking system implemented in MATLAB without requiring specialized toolboxes, ensuring accessibility and reproducibility.
- 2) A chirp preamble synchronization mechanism achieving reliable temporal alignment with peak-to-sidelobe ratios of 8.4–29.4 dB in acoustically degraded recordings.
- 3) Comprehensive empirical evaluation under realistic consumer hardware conditions with 15 independent acoustic trials across three distances.
- 4) Documentation of methodology evolution from QIM-based to spread spectrum embedding, providing practical design insights.
- 5) Quantitative identification of consumer hardware limitations (AGC, noise cancellation, frequency response) as primary barriers to acoustic watermark recovery.

E. Paper Organization

Section II reviews related work. Section III describes the proposed methodology. Section IV details methodology evolution. Section V presents the experimental setup. Section VI presents results with detailed analysis and visualization. Section VII discusses challenges and limitations. Section VIII outlines future work, and Section IX concludes.

II. LITERATURE REVIEW

A. Audio Watermarking Classification

Audio watermarking techniques are broadly classified into time-domain and transform-domain methods [6]. Time-domain approaches operate directly on audio samples and include Least Significant Bit (LSB) modification [7], echo hiding [8], and patchwork methods [2]. LSB methods replace the least significant bits of audio samples with watermark data, offering high capacity but minimal robustness. Echo hiding embeds data by introducing controlled echoes at specific delays, providing moderate robustness but limited capacity. Patchwork methods divide the signal into patches and modify statistical properties between them [9].

Transform-domain methods embed watermarks in the spectral representation, leveraging energy compaction and perceptual masking properties. These include methods based on the Discrete Cosine Transform (DCT) [10], Discrete Wavelet Transform (DWT) [11], Discrete Fourier Transform (DFT) [12], and Singular Value Decomposition (SVD) [13]. Transform-domain methods achieve superior robustness because spectral modifications distribute across multiple time-domain samples [13].

B. DCT-Based Watermarking

The DCT has been widely adopted due to its energy compaction properties and compatibility with perceptual audio coding [14]. DCT methods divide audio into frames, compute the transform, and modify selected coefficients. The coefficient range selection involves a fundamental trade-off: low-frequency coefficients carry significant perceptual energy (modifications are audible but robust), while high-frequency coefficients carry less energy (modifications are imperceptible but fragile) [15].

Mid-frequency coefficients (typically corresponding to 500–3000 Hz) offer the best compromise for acoustic robustness. This frequency range falls within the sensitivity band of most consumer speakers and microphones while providing sufficient energy for reliable detection [23]. The DCT's real-valued output (unlike the complex DFT) simplifies coefficient modification and reduces computational complexity.

C. Spread Spectrum Watermarking

Spread spectrum techniques embed watermarks by adding a scaled pseudo-random noise sequence to the host signal [16]. The watermark is spread across wide bandwidth, providing resistance to narrowband interference. At extraction, correlation with the known PN sequence recovers the watermark while the host signal's contribution averages to near zero [17].

Cox et al. [18] demonstrated spread spectrum's inherent robustness against common signal processing operations. Malvar and Florêncio [19] proposed improved spread spectrum (ISS) that accounts for host signal interference, achieving near-optimal detection performance. The key advantage of spread spectrum over deterministic embedding (such as QIM) is its graceful degradation under noise—performance degrades gradually rather than catastrophically as channel conditions worsen.

The spread spectrum approach is particularly well-suited for acoustic channels because: (i) the distributed embedding across many coefficients provides inherent averaging against noise, (ii) the PN sequence acts as a shared secret key, and (iii) the correlation-based detection is naturally robust to amplitude scaling [18].

D. Quantization Index Modulation

QIM, introduced by Chen and Wornell [20], encodes watermark bits through quantizer selection applied to host signal features. QIM achieves theoretically optimal rate-distortion performance and supports blind extraction. However, QIM is sensitive to amplitude scaling and additive noise [21], making it vulnerable to acoustic distortion. The Scalar Costa Scheme (SCS) [21] extends QIM with dithered quantization for improved security, but inherits QIM's sensitivity to channel amplitude variations.

E. Acoustic Air-Gap Watermarking Studies

Research on acoustic air-gap watermarking remains limited. Xiang and Huang [22] demonstrated watermark survival through acoustic channels using DWT-based embedding with

synchronization codes. Steinebach [23] investigated robustness against analog re-recording, finding mid-frequency DCT embedding offered the best trade-off. Arnold [24] provided evaluation frameworks for acoustic transmission conditions, emphasizing synchronization's critical role.

Most existing studies employ professional equipment (studio microphones, reference monitors) in acoustically treated environments. This creates a significant gap in understanding performance under consumer hardware conditions—the scenario most relevant to practical deployment where content monitoring occurs using commodity devices.

F. Synchronization Techniques

Temporal synchronization is the most critical challenge in acoustic watermarking [25]. Without precise alignment, frame-based extraction cannot locate correct spectral coefficients. Approaches include full-signal cross-correlation [26], pilot tone insertion [27], synchronization codes [28], and preamble-based methods.

Linear chirp signals offer excellent autocorrelation properties—sharp mainlobe peaks with low sidelobes—making them ideal synchronization preambles [29]. Chirp signals are widely used in radar and sonar systems precisely because their matched filter output produces an unambiguous peak even in noisy environments. This property translates directly to acoustic watermarking, where the chirp must be detected in a recording corrupted by speaker/microphone distortion, ambient noise, and room reverberation.

III. PROPOSED METHODOLOGY

A. System Overview

The proposed system comprises four stages: (1) watermark generation with error-correcting coding, (2) DCT-based spread spectrum embedding with chirp preamble, (3) acoustic transmission, and (4) chirp-synchronized extraction with majority-vote decoding.

Embedding Pipeline:

Host Audio → Frame Segmentation → DCT Transform → PN Sequence Generation → Spread Spectrum Embedding → IDCT → Prepend Chirp Preamble → Watermarked Audio Output

Acoustic Channel:

Speaker Playback → Air Gap (1/3/5 ft) → Microphone Recording

Extraction Pipeline:

Recorded Audio → Chirp Matched Filter Sync → Frame Segmentation → DCT Transform → PN Correlation → Sign-Based Bit Decision → Majority Vote Decoding → Extracted Watermark

Fig. 1. Complete system pipeline showing embedding, acoustic channel transmission, and extraction stages with chirp-based synchronization.

B. Watermark Generation and Error Protection

The watermark payload consists of $W = 32$ binary bits generated deterministically from a seed value using a pseudo-random number generator (PRNG). The seed acts as a shared

secret between embedder and extractor. Deterministic generation ensures reproducibility and eliminates the need to transmit the watermark through a separate channel.

To provide error resilience against channel noise, repetition coding with factor $R = 7$ is applied:

$$b_{\text{coded}}[k] = b_{\text{original}}[\lfloor k/R \rfloor], \quad k = 0, 1, \dots, N_{\text{coded}} - 1 \quad (1)$$

This produces $N_{\text{coded}} = W \times R = 32 \times 7 = 224$ coded bits. Each payload bit is independently repeated 7 times. At extraction, majority voting across each group of 7 coded bits determines the decoded bit, tolerating up to $\lfloor R/2 \rfloor = 3$ errors per group. The decoded BER as a function of coded BER p follows the binomial CDF:

$$\text{BER}_{\text{decoded}} = \sum_{k=4}^7 \binom{7}{k} p^k (1-p)^{7-k} \quad (2)$$

For example, a coded BER of 0.30 yields a decoded BER of approximately 0.126, while a coded BER of 0.20 yields approximately 0.033.

C. DCT-Based Spread Spectrum Embedding

1) *Frame Segmentation*: The host audio signal $x[n]$ of length L at sampling frequency f_s is divided into N_{coded} non-overlapping frames of length $M = 4096$ samples:

$$x_i[n] = x[(i-1) \cdot M + n], \quad n = 0, \dots, M-1 \quad (3)$$

At $f_s = 44100$ Hz, each frame spans approximately 92.9 ms. The minimum required audio duration is:

$$T_{\min} = \frac{N_{\text{coded}} \times M}{f_s} = \frac{224 \times 4096}{44100} \approx 20.81 \text{ seconds} \quad (4)$$

The frame length of 4096 was chosen as a power of 2 for computational efficiency and represents a balance between frequency resolution (each DCT bin spans $f_s/(2M) \approx 5.38$ Hz) and temporal resolution.

2) *DCT Transform*: For each frame, the Type-II DCT is computed:

$$C_i[k] = \sqrt{\frac{2}{M}} \varepsilon_k \sum_{n=0}^{M-1} x_i[n] \cos\left(\frac{\pi(2n+1)k}{2M}\right) \quad (5)$$

where $\varepsilon_k = 1/\sqrt{2}$ for $k = 0$ and $\varepsilon_k = 1$ otherwise. The DCT provides a real-valued frequency decomposition with excellent energy compaction, concentrating signal energy in lower-indexed coefficients.

3) *Coefficient Selection and PN Sequence Generation*: Embedding targets DCT bins $[k_{\text{start}}, k_{\text{end}}] = [100, 500]$, corresponding to:

$$f = \frac{k \cdot f_s}{2M} \implies 538 \text{ Hz to } 2691 \text{ Hz} \quad (6)$$

This mid-low frequency range was selected based on two criteria: (i) frequencies below 500 Hz are perceptually critical and modifications would be audible, (ii) frequencies above 3000 Hz are poorly reproduced by consumer phone speakers and poorly captured by laptop microphones. The 538–2691 Hz

range represents the intersection of robust speaker reproduction, reliable microphone capture, and acceptable perceptual impact.

For each frame i , a deterministic PN sequence $p_i[j] \in \{-1, +1\}$ of length $N_c = k_{\text{end}} - k_{\text{start}} + 1 = 401$ is generated using seed $(i + 1000)$:

$$p_i[j] = 2 \cdot \text{Bernoulli}(0.5) - 1, \quad j = 0, \dots, N_c - 1 \quad (7)$$

The deterministic seeding ensures identical PN sequences at both embedder and extractor without requiring sequence transmission.

4) *Adaptive Strength Calculation*: The embedding strength adapts to local signal energy:

$$\sigma_i = \frac{1}{N_c} \sum_{j=0}^{N_c-1} |C_i[k_{\text{start}} + j]|, \quad s_i = \alpha \cdot \sigma_i \quad (8)$$

where $\alpha = 0.5$ is the embedding strength parameter. Adaptive scaling ensures proportional watermark energy across frames with varying content, maintaining consistent imperceptibility.

5) *Spread Spectrum Modulation*: Each coded bit $b_{\text{coded}}[i]$ is embedded by modulating the PN sequence:

$$\tilde{C}_i[k_{\text{start}} + j] = C_i[k_{\text{start}} + j] + (2b_{\text{coded}}[i] - 1) \cdot s_i \cdot p_i[j] \quad (9)$$

For bit 1, the scaled PN sequence is added; for bit 0, it is subtracted. The modified coefficients are transformed back via IDCT, and sample-wise clipping to $[-1, +1]$ prevents overflow.

D. Chirp Preamble Synchronization

1) *Chirp Signal Design*: A linear frequency-modulated chirp sweeping from $f_0 = 500$ Hz to $f_1 = 4000$ Hz over duration $T_{\text{chirp}} = 1.0$ s is generated:

$$s_{\text{chirp}}(t) = A \sin\left(2\pi \left[f_0 t + \frac{f_1 - f_0}{2T_{\text{chirp}}} t^2\right]\right) \quad (10)$$

where $A = 0.8$. A 10 ms raised-cosine fade is applied at both ends to prevent spectral splatter. The frequency range was chosen to lie within the reliable bandwidth of consumer speakers (typically 300–8000 Hz) and laptop microphones (typically 100–8000 Hz).

2) *Transmitted Signal Structure*: The final signal concatenates three components:

$$y[n] = [s_{\text{chirp}}[n]] \cdot [0_{\text{gap}}[n]] \cdot [\tilde{x}[n]] \quad (11)$$

consisting of: Chirp (1.0 s) $\rightarrow$ Silence Gap (0.3 s) $\rightarrow$ Watermarked Audio (≈ 20.8 s). The gap provides an unambiguous boundary between synchronization preamble and payload.

3) *Matched Filter Synchronization*: At the receiver, the chirp is detected via frequency-domain cross-correlation:

$$R[\tau] = \text{IFFT}(\text{FFT}(s_{\text{chirp}})^* \cdot \text{FFT}(y_{\text{rec}})) \quad (12)$$

The peak location $\tau^* = \arg \max_{\tau} |R[\tau]|$ identifies the chirp start. The watermark audio begins at $n_{\text{start}} = \tau^* + M_{\text{chirp}} +$

M_{gap} . Synchronization quality is measured by the peak-to-sidelobe ratio:

$$\text{PSR} = 20 \log_{10} \left(\frac{|R[\tau^*]|}{\max_{|\tau - \tau^*| > \delta} |R[\tau]|} \right) \text{ dB} \quad (13)$$

where δ defines an exclusion zone around the main peak. PSR values above 3 dB indicate reliable synchronization.

E. Watermark Extraction and Decoding

For each aligned frame, the DCT is computed and the same PN sequence regenerated. The correlation determines the bit:

$$\rho_i = \sum_{j=0}^{N_c-1} \tilde{C}_i[k_{\text{start}} + j] \cdot p_i[j] \quad (14)$$

$$\hat{b}_{\text{coded}}[i] = \begin{cases} 1 & \text{if } \rho_i > 0 \\ 0 & \text{otherwise} \end{cases} \quad (15)$$

The host signal component averages to near zero (being uncorrelated with the PN sequence), while the watermark component produces consistent positive or negative correlation. The extraction confidence is:

$$\text{conf}_i = \min\left(1, \frac{|\rho_i|}{0.05 \cdot E_i}\right) \quad (16)$$

where $E_i = \sum |C_i[k]|$ is the coefficient energy. Majority vote decoding recovers payload bits from each group of $R = 7$ coded bits.

F. Performance Metrics

The Bit Error Rate quantifies extraction accuracy:

$$\text{BER} = \frac{1}{W} \sum_{i=0}^{W-1} b_{\text{original}}[i] \oplus \hat{b}_{\text{original}}[i] \quad (17)$$

The embedding Signal-to-Noise Ratio measures imperceptibility:

$$\text{SNR} = 10 \log_{10} \frac{\sum x^2[n]}{\sum (x[n] - \tilde{x}[n])^2} \text{ dB} \quad (18)$$

IV. METHODOLOGY EVOLUTION

A significant contribution of this work is documenting the iterative refinement process through three distinct phases, each addressing failures identified in the previous approach.

A. Phase 1: QIM-Based Embedding

The initial system used Quantization Index Modulation in the DCT domain. For each bit, 12 coefficients were randomly selected from DCT bins 200–800 (≈ 1076 – 4307 Hz at 44100 Hz). The quantization step was $\delta = \alpha \cdot \sigma_{\text{local}}$ with $\alpha = 0.08$. Two interleaved lattices encoded bits: bit 0 mapped to $\delta \lfloor c/\delta \rfloor + \delta/4$, bit 1 to $\delta \lfloor c/\delta \rfloor + 3\delta/4$.

Digital: BER = 0.0000. **Acoustic**: BER ≈ 0.50 across all distances—equivalent to random guessing. The QIM approach failed because: (i) the quantization lattice structure was destroyed by the channel's amplitude scaling, frequency-dependent gain, and nonlinear distortion, and (ii) only 12 coefficients per bit provided insufficient statistical averaging against channel noise. QIM's fundamental assumption—that coefficient amplitudes are preserved through the channel—is violated by the acoustic path.

B. Phase 2: Spread Spectrum without Chirp Sync

Recognizing QIM's vulnerability, the system was redesigned using spread spectrum modulation with several key changes: replacing QIM with PN correlation-based embedding/extraction, expanding from 12 to 601 coefficients per bit (DCT bins 200–800), increasing α from 0.08 to 0.50, and using full-signal cross-correlation for synchronization.

Digital: BER = 0.0000. **Acoustic:** BER $\approx$ 0.40–0.50. While marginally improved, results remained near random guessing. Detailed analysis revealed the primary failure mode: full-signal cross-correlation synchronization was unreliable because the acoustic channel distorted the signal envelope sufficiently to produce ambiguous, multi-peaked correlation functions. Without correct frame alignment, the DCT coefficients extracted from “wrong” frames produced random correlations with the PN sequences.

C. Phase 3: Chirp-Synchronized Spread Spectrum

The final system introduced two critical improvements:

- 1) **Chirp preamble synchronization:** A 1-second linear chirp (500–4000 Hz) replaced full-signal cross-correlation. Chirp signals provide sharp, unambiguous correlation peaks even in noisy environments.
- 2) **Lower frequency embedding:** DCT bins shifted from 200–800 to 100–500 (538–2691 Hz), targeting frequencies more reliably reproduced by consumer speakers and captured by laptop microphones.

Digital: BER = 0.0000, SNR = 23.09 dB. **Acoustic:** Mean BER of 0.40 at 1 ft and 3 ft, 0.4375 at 5 ft. The chirp synchronization achieved PSR of 8.4–29.4 dB, confirming reliable alignment. Best individual trial BER reached 0.2812 (72% correct bits).

D. Key Insights from Evolution

The three-phase evolution yielded several important findings: (1) QIM is fundamentally unsuitable for acoustic channels, (2) spread spectrum provides graceful degradation, (3) synchronization precision is as important as embedding robustness, (4) consumer hardware introduces processing artifacts absent from professional equipment, and (5) the acoustic channel's cumulative distortions are more severe than any single component.

V. EXPERIMENTAL SETUP

A. Hardware Configuration

The apparatus consisted of consumer-grade equipment: a smartphone speaker for playback, a laptop with built-in microphone for recording, Audacity software (44100 Hz, 16-bit mono WAV), and MATLAB for processing. No external speakers, amplifiers, or dedicated microphones were used, establishing a worst-case consumer baseline.

B. Audio Material

The host audio was a royalty-free instrumental track (“WavFree.wav”) at 44100 Hz, 16-bit, approximately 30 seconds, with rich harmonic content across the mid-frequency range. The audio was normalized to 95% of full scale before embedding.

C. System Parameters

Table I summarizes the system configuration.

TABLE I
SYSTEM CONFIGURATION PARAMETERS

Parameter	Value
Payload size (W)	32 bits
Repetition factor (R)	7
Coded bits (N_{coded})	224
Frame length (M)	4096 samples (92.9 ms)
DCT coefficient range	Bins 100–500
Frequency range	538–2691 Hz
Coefficients per bit (N_c)	401
Embedding strength (α)	0.50
Chirp duration	1.0 s (500–4000 Hz)
Silence gap	0.3 s
Watermark audio duration	$\approx$ 20.81 s

D. Test Protocol

Experiments were conducted in a standard indoor room without acoustic treatment. The watermarked audio (including chirp preamble) was played from the smartphone, and recording was initiated on the laptop before playback. Five independent trials were conducted at each of three distances: 1 ft ($\approx$ 30 cm), 3 ft ($\approx$ 91 cm), and 5 ft ($\approx$ 152 cm), yielding 15 total recordings. Speaker volume was set to approximately 75%.

E. Software Implementation

The complete system was implemented across nine MATLAB scripts: generate_watermark.m (payload generation), embed_watermark.m (DCT embedding), extract_watermark.m (correlation extraction), synchronize_signals.m (chirp sync), decode_watermark.m (majority voting), compute_ber.m, compute_snr.m, run_digital_test.m (baseline), and analyze_recordings.m (offline analysis). All scripts use only built-in MATLAB functions.

VI. RESULTS AND DISCUSSION

A. Digital Baseline Verification

The digital baseline confirmed perfect watermark recovery (Table II).

The SNR of 23.09 dB indicates acceptable imperceptibility, with watermark noise below the perceptual threshold for typical listening. The near-perfect confidence confirms strong PN correlation in the digital domain.

TABLE II
DIGITAL BASELINE RESULTS

Metric	Value
Bit Error Rate	0.0000 (0/32 bits)
Embedding SNR	23.09 dB
Mean Confidence	0.995
Original Watermark	01110001110110000100100001011010
Extracted Watermark	01110001110110000100100001011010

B. Acoustic Air-Gap Results

Table III presents all 15 acoustic trial results.

C. Experimental Results Visualization

Three key visualizations were generated to analyze system performance across acoustic conditions.

1) *BER vs. Distance Curve*: Fig. 2 presents the mean BER as a function of microphone distance with error bars representing one standard deviation. The plot clearly shows the dramatic transition from perfect digital extraction ($BER = 0$) to degraded acoustic performance ($BER \approx 0.40$) at the first measurement point. The horizontal dashed lines indicate the reliability threshold ($BER = 0.1$) and random guessing baseline ($BER = 0.5$).

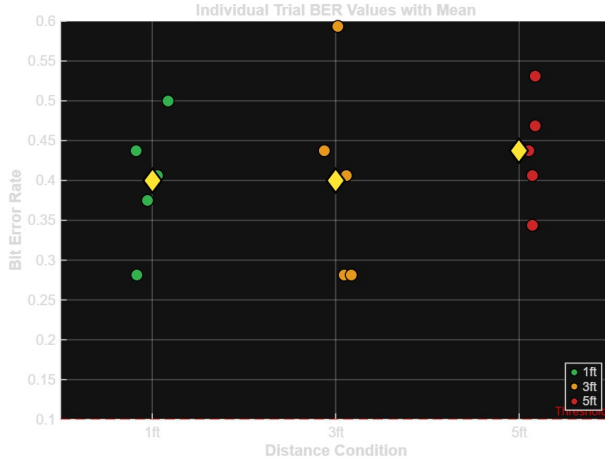


Fig. 2. Mean BER vs. microphone distance with $\pm 1\sigma$ error bars. The plot shows digital baseline ($BER = 0$), reliability threshold ($BER = 0.1$, dashed orange), and random guess level ($BER = 0.5$, dash-dot gray). All acoustic BER values fall between the threshold and random baseline, indicating partial but incomplete watermark recovery.

2) *Individual Trial Distribution*: Fig. 3 displays all 15 individual trial BER values as a scatter plot with mean diamonds. This visualization reveals the trial-to-trial variability at each distance and shows that: (i) the best-case BER of 0.2812 was achieved at both 1 ft and 3 ft, (ii) the worst-case BER of 0.5938 occurred at 3 ft Trial 5, and (iii) the 5 ft distribution is more tightly clustered than 1 ft or 3 ft distributions.

3) *Waveform Comparison*: Fig. 4 shows time-aligned 20 ms segments comparing the watermarked signal with recorded signals at each distance. The visual correlation between waveforms confirms that chirp-based synchronization achieves correct temporal alignment, while the progressive amplitude reduction and loss of fine structure with increasing distance

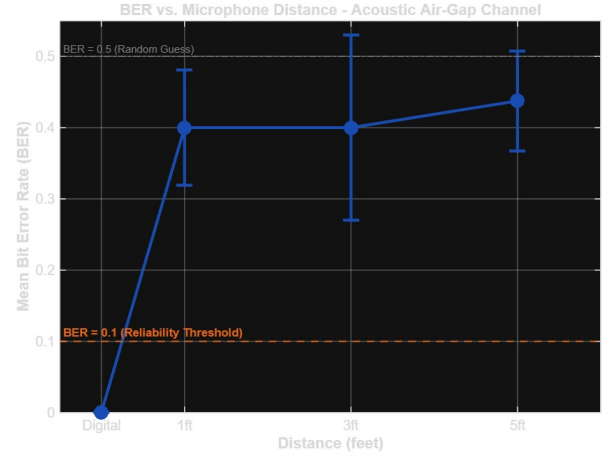


Fig. 3. Individual trial BER values across all 15 recordings. Green circles: 1 ft trials, orange circles: 3 ft trials, red circles: 5 ft trials. Yellow diamonds indicate group means. The red dashed line marks the reliability threshold.

reduction and loss of high-frequency detail with distance illustrate the increasing severity of acoustic channel distortion.

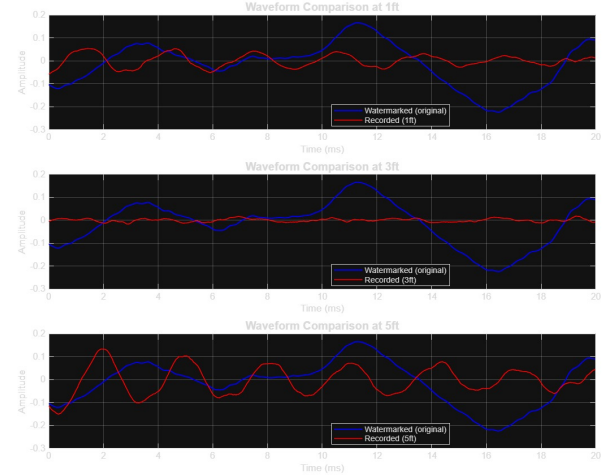


Fig. 4. Waveform comparison at 1 ft (top), 3 ft (middle), and 5 ft (bottom). Blue: watermarked original, red: recorded signal after chirp synchronization. Note the progressive amplitude reduction and loss of fine structure with increasing distance.

D. BER vs. Distance Analysis

The most significant degradation occurs between digital ($BER = 0$) and 1 ft ($BER = 0.40$)—a complete transition from perfect to severely degraded extraction. This 0.40 jump is attributable to the entire acoustic chain rather than distance alone.

A slight upward trend from 1 ft (0.4000) to 5 ft (0.4375) shows BER increasing by only 0.0375 over 4 feet. The identical results at 1 ft and 3 ft (both 0.4000) suggest that hardware processing—not propagation loss—dominates the degradation.

The 3 ft condition shows highest variance ($\sigma = 0.1296$), driven by one outlier ($BER = 0.5938$). The 5 ft condition

TABLE III
COMPLETE ACOUSTIC AIR-GAP EXPERIMENTAL RESULTS

Distance	Trial	BER	Errors/32	Ch. SNR (dB)	Offset (ms)	PSR (dB)	Confidence
Digital	—	0.0000	0	23.09	—	—	0.995
1 ft	1	0.3750	12	−2.45	2351.4	8.4	0.886
1 ft	2	0.4062	13	−1.66	1645.0	29.4	0.879
1 ft	3	0.2812	9	−2.24	1992.0	24.9	0.844
1 ft	4	0.5000	16	−1.82	1849.7	28.0	0.813
1 ft	5	0.4375	14	−2.14	2021.6	24.8	0.863
Mean	—	0.4000	12.8	−2.06	1971.9	23.1	0.857
Std	—	±0.0809	—	±0.32	±259.2	—	—
3 ft	1	0.2812	9	−2.63	1558.2	21.7	0.898
3 ft	2	0.2812	9	−1.66	1884.2	23.0	0.870
3 ft	3	0.4375	14	−2.11	1575.1	22.6	0.890
3 ft	4	0.4062	13	−2.01	1945.0	24.2	0.895
3 ft	5	0.5938	19	−1.89	1782.5	24.7	0.860
Mean	—	0.4000	12.8	−2.06	1749.0	23.2	0.883
Std	—	±0.1296	—	±0.36	±176.4	—	—
5 ft	1	0.4688	15	−1.70	1775.3	20.9	0.884
5 ft	2	0.5312	17	−3.13	1982.7	14.5	0.863
5 ft	3	0.4062	13	−2.80	1954.9	17.6	0.846
5 ft	4	0.3438	11	−2.17	1684.1	16.0	0.891
5 ft	5	0.4375	14	−2.97	2002.4	17.0	0.857
Mean	—	0.4375	14.0	−2.55	1879.9	17.2	0.868
Std	—	±0.0699	—	±0.60	±141.9	—	—

shows lowest variance ($\sigma = 0.0699$), suggesting the SNR has stabilized at a consistently low level where variability is reduced.

The minimum BER of 0.2812 (9/32 errors, 72% accuracy) was achieved in three trials, indicating partial watermark information survives the acoustic channel under favorable conditions.

E. Synchronization Performance Analysis

TABLE IV
CHIRP SYNCHRONIZATION PERFORMANCE

Metric	1 ft	3 ft	5 ft
Mean PSR (dB)	23.1	23.2	17.2
Min PSR (dB)	8.4	21.7	14.5
Max PSR (dB)	29.4	24.7	20.9
Mean Offset (ms)	1971.9	1749.0	1879.9

All PSR values exceed the 3 dB threshold (Table IV), confirming reliable chirp detection. The decreasing PSR with distance (23.1 $\rightarrow$ 23.2 $\rightarrow$ 17.2 dB) reflects reduced SNR at greater distances. The offset variation (141–259 ms std) reflects user-dependent playback timing, not synchronization errors.

F. Channel SNR Analysis

Channel SNR values of −2.06 to −2.55 dB indicate noise power exceeds signal power—consistent with severe consumer hardware distortion. The identical SNR at 1 ft and 3 ft (−2.06 dB) confirms hardware processing as the dominant noise source. The slight decrease at 5 ft (−2.55 dB) adds distance-dependent attenuation and ambient noise.

G. Comparison with Random Guessing

A BER of 0.5 represents random guessing. Observed values (0.40–0.4375) are consistently below this baseline. Using the binomial distribution, $P(X \leq 12.8 \mid n = 32, p = 0.5) \approx 0.133$. The consistency across 15 trials and the upward BER trend with distance provide evidence of partial (though insufficient) watermark detection.

H. Summary of Results

TABLE V
SUMMARY OF KEY FINDINGS

Condition	Mean BER	Channel SNR
Digital Baseline	0.0000	23.09 dB
1 ft Acoustic	0.4000 $\pm$ 0.0809	−2.06 dB
3 ft Acoustic	0.4000 $\pm$ 0.1296	−2.06 dB
5 ft Acoustic	0.4375 $\pm$ 0.0699	−2.55 dB

VII. CHALLENGES AND LIMITATIONS

A. Consumer Hardware Constraints

Four hardware factors primarily degraded acoustic performance:

Automatic Gain Control (AGC): Laptop microphones automatically adjust gain levels, introducing time-varying amplitude modifications that distort watermark energy relationships across DCT coefficients. AGC changes the relative amplitudes of coefficients within and across frames, directly affecting the PN correlation computation.

Noise Cancellation: Modern laptops employ digital noise cancellation algorithms that actively modify the captured

spectrum. These algorithms are designed to suppress non-speech signals, potentially attenuating the exact frequency bands (538–2691 Hz) used for watermark embedding.

Speaker Frequency Response: Mobile phone speakers exhibit limited and uneven frequency response, with significant roll-off below 500 Hz and resonance peaks that cause non-uniform attenuation of watermark energy across different DCT coefficients.

Microphone Limitations: Built-in laptop microphones are optimized for speech capture (300–3400 Hz) with directional characteristics designed for near-field voice input, providing reduced sensitivity for far-field acoustic watermark signals.

B. Acoustic Environment

Room reflections create multipath propagation, causing temporal smearing that spreads each frame’s energy into adjacent frames. This inter-frame interference disrupts the coefficient structure assumed by the extractor. The experiments were conducted without acoustic treatment, representing realistic but challenging conditions.

C. Clock Drift

Independent clock oscillators in the playback and recording devices introduce progressive temporal misalignment. Over the 20.81-second watermark duration, even 50 ppm clock difference accumulates to approximately 1 ms of drift, causing increasing frame boundary errors for later frames. This cumulative effect means that even with perfect initial synchronization, later frames experience progressively worse alignment.

D. Signal Attenuation with Distance

The recorded signal RMS decreased from 0.1324 at 1 ft to 0.0314 at 5 ft—a $4.2\times$ reduction. While RMS normalization compensates for overall level differences, the reduced signal-to-ambient-noise ratio fundamentally limits extraction accuracy. At 5 ft, ambient noise constitutes a larger fraction of the captured signal, diluting the watermark energy.

VIII. FUTURE WORK

A. Professional Hardware Evaluation

External condenser microphones with flat frequency response and no AGC, combined with dedicated speakers with known frequency characteristics, could substantially reduce acoustic BER. We hypothesize $\text{BER} < 0.10$ is achievable at 1 ft with professional hardware, as our analysis identified hardware processing as the dominant degradation source.

B. Advanced Error Correction Coding

Replacing repetition coding ($R = 7$) with capacity-approaching codes could dramatically improve decoded BER: LDPC codes approaching Shannon limit performance, turbo codes with iterative decoding, or convolutional codes with soft-decision Viterbi decoding that exploits correlation confidence values. With observed coded BER of approximately 0.40, a rate-1/3 turbo code could potentially achieve decoded BER below 0.05.

C. Adaptive Frequency Selection

Channel-aware embedding that estimates the acoustic channel frequency response and concentrates watermark energy in least-attenuated bands could improve robustness. A training phase using the chirp preamble itself could characterize the channel for subsequent coefficient selection.

D. Perceptual Masking Integration

Psychoacoustic masking models (as used in MP3/AAC encoders) would enable higher embedding strength in perceptually masked regions without increasing audibility, potentially improving watermark-to-noise ratio by 10–15 dB.

E. Expanded Experimental Scope

Future experiments should include: additional distances (0.5, 2, 4, 7, 10 ft), multiple room types (anechoic, office, lecture hall), various noise conditions, and cross-hardware evaluation with multiple speaker and microphone models.

IX. CONCLUSION

This paper presented a DCT-based spread spectrum audio watermarking system with chirp-synchronized extraction for acoustic air-gap channels. The system was implemented entirely in MATLAB using built-in functions, ensuring reproducibility without specialized toolboxes.

The digital baseline confirmed a fully functional pipeline with perfect recovery ($\text{BER} = 0.0000$) and acceptable imperceptibility ($\text{SNR} = 23.09$ dB). The 32-bit payload, protected by $R = 7$ repetition coding, was embedded via spread spectrum modulation in DCT bins 100–500 (538–2691 Hz).

Acoustic experiments at 1, 3, and 5 feet using consumer hardware yielded mean BER values of 0.4000, 0.4000, and 0.4375 respectively. While exceeding the 0.10 reliability threshold, these values are consistently below the 0.50 random guessing baseline, indicating partial watermark recovery. The chirp synchronization mechanism achieved PSR of 8.4–29.4 dB, confirming robust temporal alignment even through severely degraded acoustic channels.

The documented methodology evolution from QIM ($\text{BER} \approx 0.50$) through spread spectrum with full-signal synchronization ($\text{BER} \approx 0.45$) to chirp-synchronized spread spectrum ($\text{BER} \approx 0.40$) demonstrated incremental improvements, with each refinement addressing specific failure modes.

The key finding is that consumer hardware processing (AGC, noise cancellation, frequency response limitations) constitutes the primary barrier to reliable acoustic watermark recovery, rather than algorithmic limitations or acoustic propagation effects. The system’s perfect digital-domain performance confirms algorithmic correctness, while the acoustic-domain challenges establish a quantitative baseline for future improvements with professional equipment, advanced error correction, and adaptive embedding strategies.

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